Unit: Unit 0: Foundations of Algebra
Topic: Inequalities

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Solve the inequality $2x + 5 > 11$
(A) $x > 3$
(B) $x < 3$
(C) $x \geq 3$
(D) $x \leq 3$
(E) $x = 3$

Q2. Graph the solution to $x - 2 < 7$ on a number line
(A) $(-\infty, 9)$
(B) $(-\infty, 5)$
(C) $(-\infty, 9]$
(D) $[5, \infty)$
(E) $(-5, \infty)$

Q3. Write the solution to $4x > 12$ in interval notation
(A) $(-\infty, 3)$
(B) $(-\infty, -3)$
(C) $(-3, \infty)$
(D) $(-\infty, -3]$
(E) $(3, \infty)$

Q4. Solve the inequality $x/3 + 2 > 5$
(A) $x > 9$
(B) $x < 9$
(C) $x \geq 9$
(D) $x \leq 9$
(E) $x = 9$

Q5. Graph the solution to $2x + 1 \leq 9$ on a number line
(A) $(-\infty, 4)$
(B) $(-\infty, 4]$
(C) $[4, \infty)$
(D) $(-\infty, -4]$
(E) $(-4, \infty)$

Q6. Write the solution to $x - 4 > 2$ in interval notation
(A) $(-\infty, 6)$
(B) $(-\infty, 6]$
(C) $[6, \infty)$
(D) $(-6, \infty)$
(E) $(6, \infty)$

Q7. Solve the inequality $3x - 2 < 14$
(A) $x < 16/3$
(B) $x > 16/3$
(C) $x \geq 16/3$
(D) $x \leq 16/3$
(E) $x = 16/3$

Q8. Graph the solution to $x/2 + 1 > 3$ on a number line
(A) $(-\infty, 4)$
(B) $(-\infty, 4]$
(C) $[4, \infty)$
(D) $(-4, \infty)$
(E) $(4, \infty)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $2x + 5 > 3x - 2$ and write the solution in interval notation

Q10. Graph the solution to $x^2 - 4x - 3 > 0$ on a number line and write the solution in interval notation

Chef's Tips

- Check for any calculations or algebraic manipulations
- Verify that the solutions are in the correct format
- Ensure that students provide clear and concise explanations for their answers